Technical University of Crete

School of Electrical and Computer Engineering

Telecommunication Systems I

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Exercise 1

Submission date: 16 March 2026

The assignment may be submitted by groups of $\leq$ two people

Points: 100/1000

"Plagiarism" or "similarities beyond what is normal"
will be penalized with a zero grade.

The weight of each question is given in front of it.

On the first page of the report, state the number of hours required to complete the exercise.

The report must include the full course information, the exercise number, and the names and student ID numbers of the members of the group.

The report should have the following structure:

1. Statement of the question and its points.
2. Figure (where needed)
3. MATLAB code used to produce the figure (where needed)
4. Comments or explanation (where needed).

First, we will study baseband communication with 2–PAM modulation and truncated Square Root Raised Cosine (SRRC) pulses.

The function srrc_pulse.m, which constructs truncated SRRC pulses, is provided in "Documents." Read its code carefully.

Your code must be fully parameterized, so that changing only the value of T is sufficient for the steps below to run correctly for any symbol period T.

A.1 Construct SRRC pulses $\phi(t)$. Indicative values: $T = 10^{-3}\,\text{sec}$, over $= 10$, $T_s = \frac{T}{\text{over}}$, $A = 4$, and roll-off factor $a = 0, 0.5, 1$ (here A denotes half the length of the truncated pulse measured in symbol periods).

(5) Plot the pulses on the same figure (a **common** plot) with an appropriate time axis, for $T = 10^{-2}\,\text{sec}$, over $= 10$, $A = 4$, and $a = 0, 0.5, 1$.

(5) What do you observe about the rate at which the pulse amplitude "decays" as the absolute time $|t|$ increases, with respect to the value of a? Which pulse decays faster?

A.2 Using the functions `fft` and `fftshift`, compute the Fourier transforms $\Phi(F)$ of the pulses from the previous step at N_f equally spaced frequency points over $[-\frac{F_s}{2}, \frac{F_s}{2})$ (e.g., $N_f = 1024, 2048$).

Plot the energy spectral density $|\Phi(F)|^2$ of these pulses

(a) (5) on a common `plot`,

(b) (5) on a common `semilogy` (note that `semilogy` lets you study the values of $|\Phi(F)|^2$ over ranges where they are very small, which cannot be done easily with a linear `plot`).

A.3 The theoretical bandwidth of infinite-duration pulses is $\mathrm{BW} = \frac{1+a}{2T}$.

(5) Compute the theoretical bandwidth for each of the three pulses.

(5) In practice, since truncated pulses have (theoretically) infinite bandwidth, a more **practical** definition of bandwidth is needed. On the common semilogy plot from question A.2, draw a horizontal line with value c (e.g., $c = \frac{T}{10^3}$) and consider values below this line to be "practically zero." Under this definition, what is the approximate bandwidth of the three pulses (the "Zoom in" feature will be helpful)? Which pulse is more bandwidth-efficient?

How does the bandwidth change if $c = \frac{T}{10^5}$? In this case, which pulse is more efficient?

If you implemented the above successfully, you should have realized that when we use truncated pulses, unlike the ideal case where we use the entire (infinite) pulse, it is not uniquely defined which pulse is optimal in terms of bandwidth. It depends on what we define as "practically zero," and choosing the "optimal" pulse requires care, experience, and experimentation.

Next, we will experiment with the (approximate) orthonormality properties of $\phi(t)$, with respect to its shifts by integer multiples of T.

B.1 For $T = 10^{-2}$, $A = 4$, $a = 0, 0.5, 1$, and $k = 0, 1, \ldots, 2A$ (you will obtain corresponding results for negative k)

1. plot $\phi(t)$ and $\phi(t - kT)$ on the same plot,
2. plot the product $\phi(t)\,\phi(t - kT)$,
3. numerically approximate the integral of the product $\phi(t)\,\phi(t - kT)$ using the method described in the notes.

(10) Include in your report the results of steps 1 and 2 for $a = 0.5$ and $k = 0, 1, 2, 3$.

(10) Include in your report the values of the integrals from step 3 for $a = 0, 0.5, 1$ and $k = 0, 1, 2, 3$, and try to explain them.

If you implemented the above successfully, you should have realized that truncated SRRC pulses are **approximately orthonormal**, with respect to shifts by kT, with the approximation improving as a approaches 1. Also, if you experiment with A, you will see that the approximation improves as A increases.

Finally, we will simulate a baseband PAM system that transmits N bits using 2-PAM modulation. Let $T = 10^{-2}$ sec, over $= 10$, $a = 0.5$, and $A = 4$ (again, A equals half the length of the truncated pulse measured in symbol periods).

C.1 Generate N bits b_i, for $i = 0, \ldots, N-1$ (e.g., $N = 50, 100$), using the command

$$\texttt{b} = (\texttt{sign}(\texttt{randn}(\texttt{N}, \texttt{1})) + \texttt{1})/\texttt{2}; \qquad (1)$$

C.2 The baseband 2-PAM system is implemented as follows.

1. (10) Write a function

   $$\texttt{X} = \texttt{bits\_to\_2PAM}(\texttt{b});$$

   which takes as input the bit sequence `b` and outputs the sequence of 2-PAM symbols `X` using the mapping:

   $$0 \longrightarrow +1,$$

   $$1 \longrightarrow -1.$$

2. (10) Simulate the signal

   $$X_\delta(t) = \sum_{k=0}^{N-1} X_k \, \delta(t - kT), \qquad (2)$$

   using the command

   $$\texttt{X\_delta} = \texttt{1}/\texttt{T\_s} * \texttt{upsample}(\texttt{X}, \texttt{over}); \qquad (3)$$

   Define the appropriate time axis and plot $X_\delta(t)$.

3. (10) Create a *truncated* SRRC pulse $\phi(t)$ (using the parameters above) and simulate the convolution $X(t) = X_\delta(t) \circledast \phi(t)$. Construct the appropriate time axis and plot $X(t)$.[1]

4. Assuming an ideal channel, the receiver input is $X(t)$.

   (10) Simulate the convolution $Z(t) = X(t) \circledast \phi(-t)$. Plot $Z(t)$ over the corresponding time axis and find the values it takes at time instants kT, for

[1] Recall that the convolution of two continuous-time signals $x_1(t)$ and $x_2(t)$ that are nonzero over the time intervals $[t^1_{\min}, t^1_{\max}]$ and $[t^2_{\min}, t^2_{\max}]$, respectively, is nonzero over the interval $[t^1_{\min}+t^2_{\min},\ t^1_{\max}+t^2_{\max}]$.

$k = 0, \ldots, N-1$. Relate these values to X_k, for $k = 0, \ldots, N-1$. Can you explain the phenomenon?

(10) A graphical way to compare $Z(kT)$ with X_k, for $k = 0, \ldots, N-1$, is `hold on` the plot of $Z(t)$ and execute

```
stem([0 : N - 1] * T, X);
```

where X is the vector of symbols X_k, $k = 0, \ldots, N-1$. What do you observe? Can you explain the phenomenon?